

Introduction to the Astrophysics Data System (ADS): Paper Scavenger Hunt

You should work together with a partner on all in class activities and have discussions, but each person should work on their own computer, and complete their own worksheet.

Learning Objectives: Use the NASA/ADS interface to search for papers using appropriate keywords.

Being able to effectively search for and find scientific information is a crucial skill for research. In this exercise we will use the Astrophysical Data System (ADS) to search for astronomy information. You will learn to understand the different keywords and their functions, as well as how to select and sort paper listings based on key information.

1. Getting started.

- Open a web browser and go to <https://ui.adsabs.harvard.edu/>. Make sure the ‘Modern Form’ tab is selected.

2. **Search syntax.** You can see there is already some documentation on how to do searches. What is the syntax to search for the following (write in below):

— a particular author:

— the first author of an article:

— an article year:

— a range of years:

— specific text in the article abstract or title:

— specific text appearing anywhere in the article:

3. **Author Search.** Let's orient our author search on Vera Rubin, who is credited with discovering dark matter.

■ Search for a list of all publications in which she is first author.

— How many do you find?

— What is the year and title of her first, first author paper? (Note you can figure this out faster by reordering the list of publications — in the menu at the top, click the little graph next to Date so that it inverts.)

4. **Limiting Searches.** Let's make sure we only have the work of only one (the right!) Vera Rubin.

■ Click on the down arrow next to her name under authors. Remove the imposter Vera Rubin (hint see the initials; you may need to click “more” to see all authors). To further simplify the list, let's look only at papers that were refereed. Look at the menu on the left and select ‘*refereed*’, then click “limit to”.

— *How many first author, refereed papers did she write?*

— *What are the total number of citations of these paper?* (Select the citations tab in the plot on the right).

5. **Collaborations.** All the authors that appear on papers with the searched author can be found in the menu on the left.

■ Look at the list of collaborators. Who did she collaborate with most frequently?

6. Abstracts and Papers.

- Find the entry for Vera Rubin's most frequently cited work as first author. You can do this by sorted the list by the most cited works. What is the publication year and title of this work?

- Click on this paper and read the abstract. Write 1-2 sentences summarizing what this paper is about.

- Links to the paper itself and any data associated with it are listed to the right of the abstract. Click the paper (PDF) icon next to ADS under FULL TEXT SOURCES (the other icon takes you to the journal website itself, from where you can also access the PDF). Skim the introduction. What is the reason that this study is novel/important?

7. **Citing papers in ADS.** If we want to cite Vera Rubin's most cited paper in our end-of-semester RNAAS writeup we need the citation information. ADS provides the citation information in a variety of common citation formation, including AASTex and BibTeX.

- Click on '*Export Citation*' on the left and choose BibTeX from the dropdown menu. This text can be copied and pasted into a bibtex file, which is readable by LaTeX (we'll discuss this more later in the semester).

8. **Searching by Topic.** Let's suppose you want to find information but you don't know anything about the topic and therefor don't have any particular author you want to search for.

- Use the *keyword* search to find the most cited paper in the field of “high-redshift galaxies.” You can find “keyword” under the “all search terms” menu on the upper-left.

— What is the paper author and title?

— Is this paper a review? If not, modify the search to find the most cited *review* on high-redshift galaxies. In astronomy such reviews are usually in the journal *Annual Reviews of Astronomy & Astrophysics* (ARA&A). You can do this by adding **and** to your search, and then including **bibstem:"ARA&A"**. What is the most-cited high-redshift review?

9. Complex Searches. Let's try to find all the refereed papers that I've written about high-redshift galaxies. How would you structure the search?

- Run the search. How many papers do you find? (Hint: you may need some logic).
- How many of these are refereed?
- How many citations does the highest-cited paper have?
- What is the most highly cited refereed paper in astronomy from an author at UT Austin over the past decade? Hint: ADS has our institution name as "U TX Austin".

10. Astronomer Andrea Ghez won the 2020 Nobel Prize in Physics. How many citations does her most influential first author paper have?